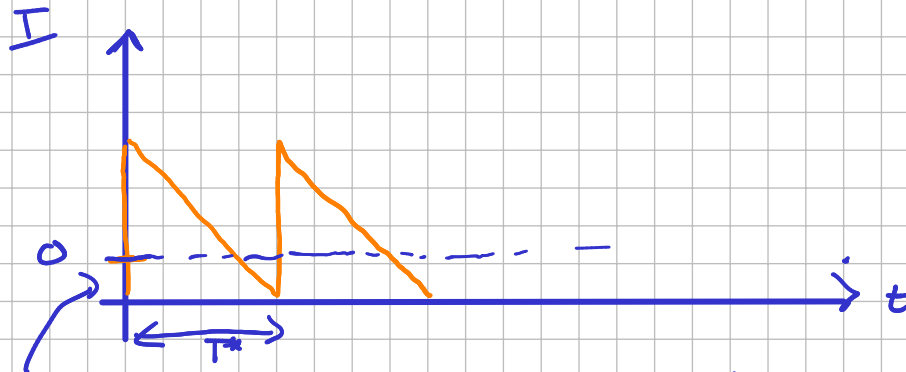


5 ...Continued

d) Determine the cycle length with the positive stock and with the backlog.



Anything below zero is backlog!

$$T^* = \frac{Q^*}{D} \frac{\text{units}}{\frac{\text{units}}{\text{year}}} = \frac{2323.790}{18000} = 0.129 \text{ years}$$

⇒ Depending on the number of working days per year, this can vary.

⇒ Assuming 300 days / year:

$$300 \frac{\text{days}}{\text{year}} \cdot 0.129 \text{ years} = 38.7 \text{ days}$$

$$T^* = 38.7 \text{ days}$$

⇒ The fraction of the cycle time for which the stock is positive is

$$t_p = \frac{I^*}{D} \text{ years}$$

⇒ The time fraction for which we have negative stock is our Backlog

$$t_n = \frac{B^*}{D} \text{ years}$$

$$\frac{t_p}{T^*} = \frac{\frac{H^*}{D}}{\frac{Q^*}{D}} = \frac{I^*}{Q^*} = \frac{Q^* - B^*}{Q^*} = \frac{2323.790 - 464.758}{2323.790}$$

$$\frac{t_p}{T^*} = 0.800$$

$\Rightarrow$ 80% of the cycle,
we have positive
stock

$$\frac{t_p}{T^*} = \frac{B^*}{Q^*} = \frac{464.758}{2323.790} = 0.2$$

$\Rightarrow$ 20% of
the cycle
we are
in the
red!

8) Economic Production Quantity (EPQ)

A factory produces filter cartridges internally with the following parameters

$$D = 30\,000 \frac{\text{units}}{\text{year}}, \quad S = 400 \frac{\text{EUR}}{\text{order}}, \quad h = 8 \frac{\text{EUR}}{\text{unit} \cdot \text{year}}$$

$$p = 90\,000 \frac{\text{units}}{\text{year}}$$

- a) The total cost function $TC(Q)$ is the sum of the setup costs and holding costs, adjusted for the production - to - consumption ratio:

$$TC(Q) = \frac{DS}{Q} + \frac{hQ}{2} \left(1 - \frac{D}{p}\right)$$

$$\begin{aligned} TC(Q) &= \frac{30\,000 \cdot 400}{Q} + \frac{8Q}{2} \left(1 - \frac{30\,000}{90\,000}\right) \\ &= \frac{12\,000\,000}{Q} + 4Q \left(\frac{2}{3}\right) \end{aligned}$$

$$TC(Q) \approx \frac{12\,000\,000}{Q} + 2.667Q$$

- b) Optimal Production Lot size Q^*

=> The optimal production lot size is determined by the 1st-order condition of the total cost function, resulting in the EPQ formula:

$$Q^* = \sqrt{\frac{2DS}{h(1-D/p)}}$$

$$2DS = 2 \cdot 30\,000 \cdot 400 = 24\,000\,000$$

$$h \left(1 - \frac{D}{p}\right) = 8 \left(1 - \frac{1}{3}\right) = \frac{16}{3} \approx 5.333$$

$$Q^* = \sqrt{\frac{24\,000\,000}{16/3}} \approx 2121.320$$

Interpretation: To minimize the cost of setup and inventory holding, the factory should produce the cartridges in batches of about 2121 units.

c) In the EPQ model, the maximum inventory $I_{\max}$ is reached at the end of the production run.

$\Rightarrow$ It is calculated by multiplying the lot size by the proportion of product that goes into storage.

$$I_{\max} = Q^* \left(1 - \frac{D}{P}\right)$$

$$\Rightarrow I_{\max} = 2121.32 \left(1 - \frac{1}{3}\right) = 2121.32 \left(\frac{2}{3}\right)$$

$$I_{\max} = 1414.21 \text{ units}$$

* Even though 2121.32 units per cycle are produced, we only store a maximum of 1414.21 units at any time because 707.11 units are shipped or consumed while the production line is still running.

d) Cycle time and Production Uptime

$\Rightarrow$ The temporal aspects of the EPQ model define the production schedule.

Cycle length

$$T^* = \frac{Q^*}{D} = \frac{2121.32}{30\,000} = 0.071 \text{ years}$$

Production Uptime is the duration of active production phase within one cycle.

$$t_p = \frac{Q^*}{P} = \frac{2121.32}{90,000} = 0.024 \text{ years}$$

12 One-stage Bullwhip Model Analysis

⇒ For a one-stage Bullwhip model, the denominator of the demand-to-order transfer function is given by

$$\tau s^2 + s + K.$$

Compare this with the standard second-order characteristic form

$$s^2 + 2\zeta\omega_n s + \omega_n^2.$$

Consider the three parameter sets:

A: $K = 0.1$, $\tau = 1$

B: $K = 0.25$, $\tau = 1$

C: $K = 1$, $\tau = 2$

a) Determine ω_n and ζ

⇒ To compare to the second-order characteristic form, we re-write the denominator of the TF by division by τ .

$$s^2 + \frac{1}{\tau}s + \frac{1}{\tau}K$$

$$\omega_n^2 = \frac{K}{\tau} \Rightarrow \omega_n = \sqrt{\frac{K}{\tau}}$$

$$2\zeta\omega_n = \frac{1}{\tau} \Rightarrow \zeta = \frac{1}{2\tau\omega_n} = \frac{1}{2\sqrt{K\tau}}$$

For case A: ($K=0.1$, $\tau=1$):

$$\omega_n = \sqrt{\frac{0.1}{1}} = 0.316$$

$$\zeta = \frac{1}{2(1)(0.316)} = 1.581 \quad (\text{Overdamped})$$

For case B: ($k=0.25$, $\tau=1$):

$$\omega_n = \sqrt{\frac{0.25}{1}} = \sqrt{0.25} = 0.500$$

$$\zeta = \frac{1}{2(1)(0.5)} = 1.000 \text{ (critically damped)}$$

For case C: ($k=1$, $\tau=2$):

$$\omega_n = \sqrt{\frac{1}{2}} = 0.707$$

$$\zeta = \frac{1}{2 \cdot 2 \cdot 0.707} \approx 0.354 \text{ (underdamped)}$$

b) System Classification

Case A: Overdamped

In the context of the Bullwhip effect, this means the order response to demand change will be slow and sluggish! It will not overshoot or oscillate!

Case B: Critically damped

System reaches the new steady-state order level in the shortest time possible without any oscillation.

Case C: Underdamped

Bullwhip! Change in demand causes order signal to overshoot and oscillate before settling.

c) Rank the cases by oscillation.

⇒ Homework